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SUB CODE: DSA0502

SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO4 ASSESSMENT TOOL 1

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Question 1 – E-Commerce Sales Analysis

An e-commerce company has Customer, Product, Order, and Payment datasets. Perform data exploration, join the datasets, analyze sales by product/category/location, identify correlations and outliers, perform time-based analysis, and create suitable visualizations. Provide 5 key insights and recommendations.

AIM: To Explain that the project analyzes an e-commerce dataset to understand sales performance, customer behavior, product categories, locations, payment methods, correlations, outliers, and sales trends.

DATASETS USED: customers , products, orders, payments , order details.

TOOLS USED: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook / VS Code.

SALES ANALYSIS:

- Total Sales
- Total Orders
- Average Order Value
- Total Customers.

```
# SALES ANALYSIS

print("===== SALES ANALYSIS =====")

# Total sales
total_sales = sales_data["payment_value"].sum()

# Number of orders
total_orders = sales_data["order_id"].nunique()

# Average order value
average_order_value = total_sales / total_orders

# Total customers
total_customers = sales_data["customer_unique_id"].nunique()

print("Total Sales:", round(total_sales, 2))
print("Total Orders:", total_orders)
print("Average Order Value:", round(average_order_value, 2))
print("Total Customers:", total_customers)
```

[10] ✓ 0.1s Python

```
... ===== SALES ANALYSIS =====
Total Sales: 16008872.12
Total Orders: 99441
Average Order Value: 160.99
Total Customers: 96096
```

PRODUCT CATEGORY ANALYSIS:

Explain which product categories generated the highest sales.

```
... ===== TOP 10 PRODUCT CATEGORIES =====
```

	product_category_name	Total_Sales	Total_Orders	Average_Price
11	beleza_saude	1297490.77	8836	130.074263
66	relogios_presentes	1253143.30	5624	202.087292
13	cama_mesa_banho	1092551.02	9417	92.408950
32	esporte_lazer	1023996.34	7720	114.476952
44	informatica_acessorios	942277.57	6689	116.589652
54	moveis_decoracao	765093.89	6449	87.499301
72	utilidades_domesticas	666587.00	5884	90.630455
26	cool_stuff	662309.49	3632	166.117254
8	automotivo	616752.51	3897	140.843231
40	ferramentas_jardim	518217.54	3518	113.296358

LOCATION ANALYSIS:

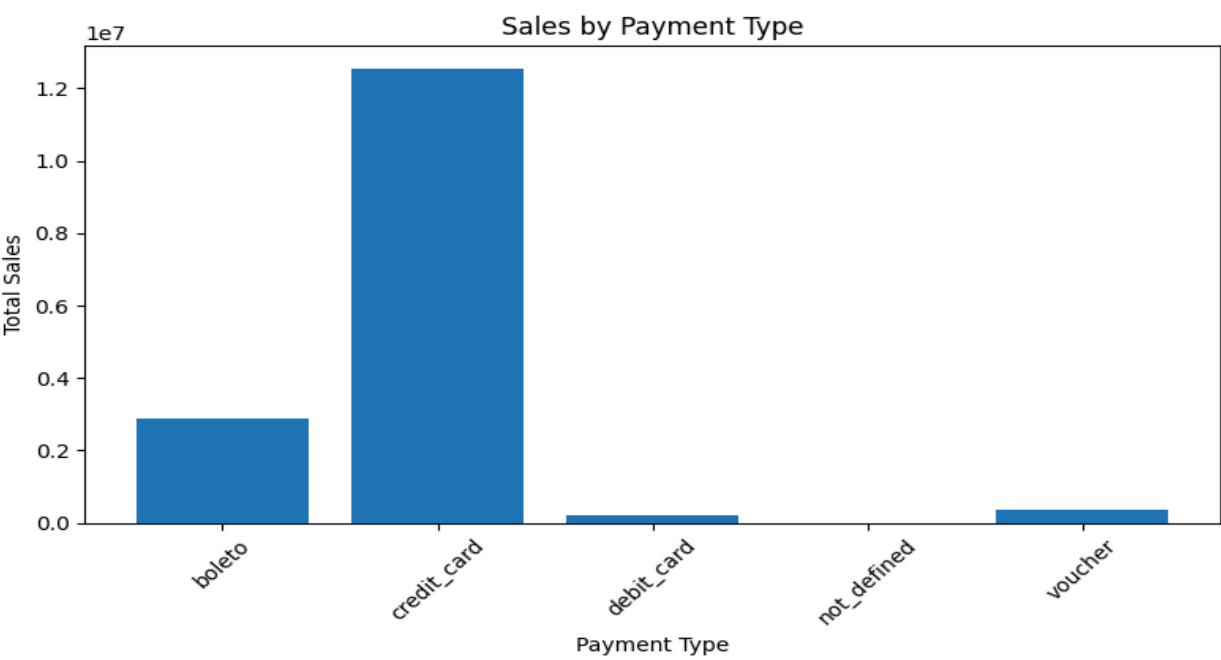
Explain sales performance by customer state and city.

...
===== TOP 10 PRODUCTS BY SALES =====
...

	product_id	Total_Sales	Total_Orders	Average_Price
24086	bb50f2e236e5eea0100680137654686c	68485.00	187	327.679426
27613	d6160fb7873f184099d9bc95e30376af	56948.83	35	1388.995854
14068	6cdd53843498f92890544667809f1595	55779.90	151	350.816981
27039	d1c427060a0f73f6b889a5c7c61f2ac4	49141.40	323	137.650980
19742	99a4788cb24856965c36a24e339b6058	46308.96	467	88.207543
4996	25c38557cf793876c5abdd5931f922db	44829.32	38	1018.848182
8051	3dd2a17168ec895c781a9191c1e95ad7	41682.20	255	149.935971
22112	aca2eb7d00ea1a7b8ebd4e68314663af	38248.20	431	71.358582
10867	53b36df67ebb7c41585e8d54d6772e08	38158.21	306	116.691774
12351	5f504b3a1c75b73d6151be81eb05bdc9	37733.90	63	598.950794

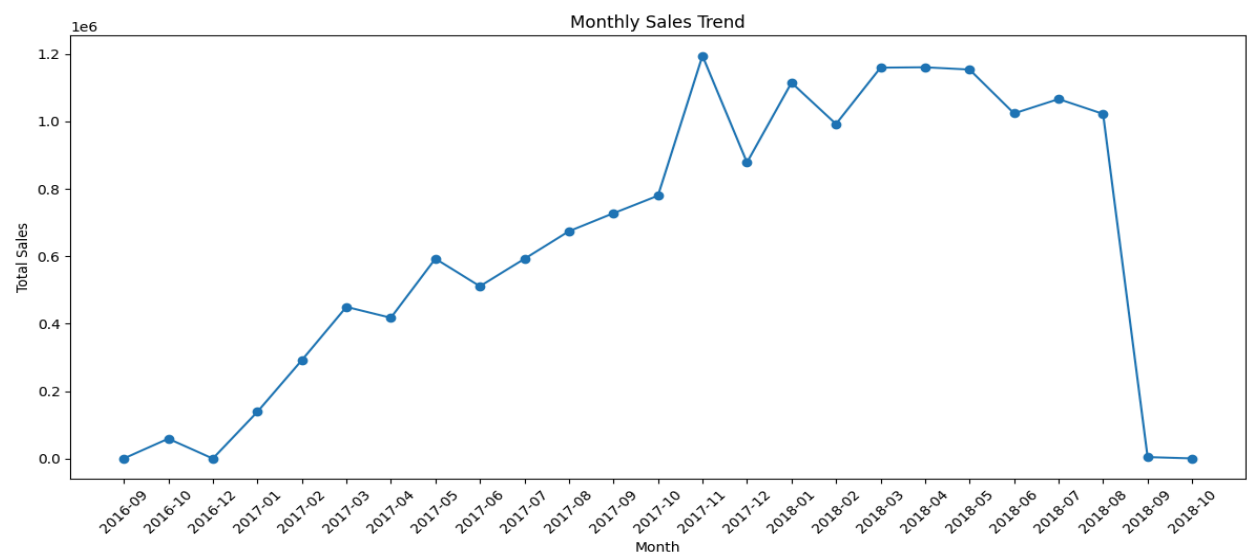
PAYMENT ANALYSIS:

Discuss the most commonly used/highest-sales payment method.



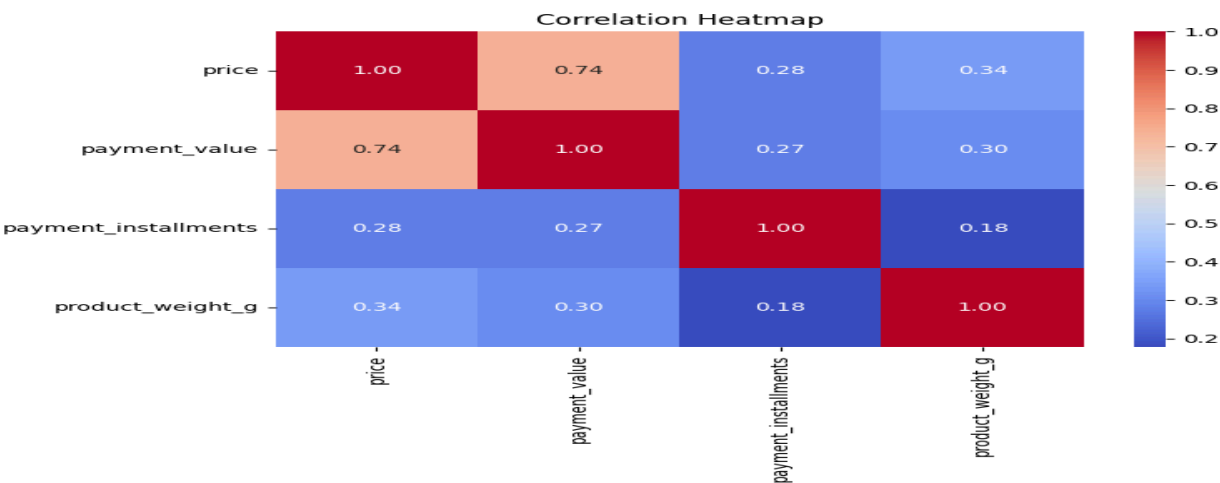
TIME-BASED ANALYSIS:

Explain monthly sales trends and identify the highest-sales month.



CORRELATION ANALYSIS:

- Product price
- Payment value
- Payment installments
- Product weight.

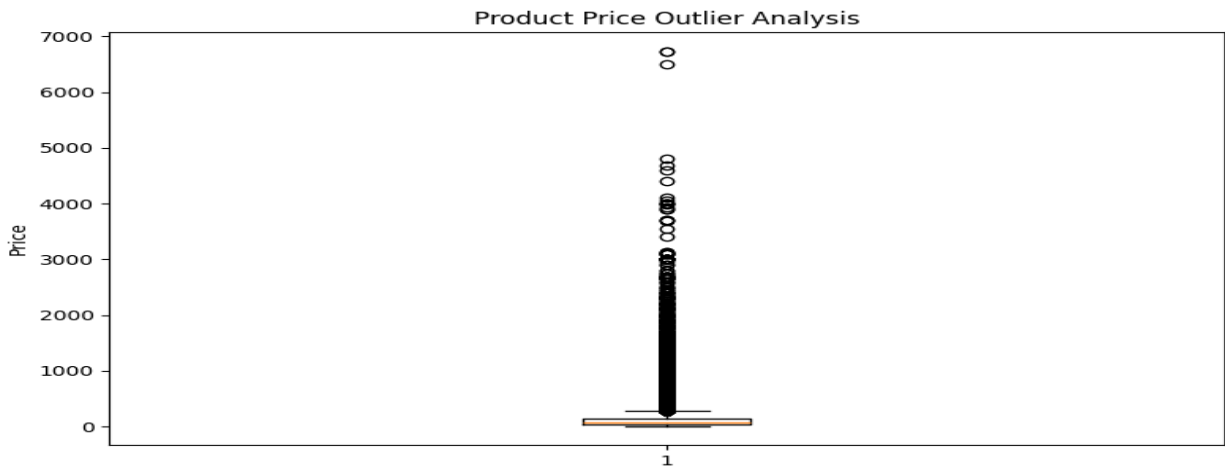


OUTLIER ANALYSIS:

Explain that the **IQR method** was used to identify unusual values.

Analyze:

- Product price
- Payment value.



FIVE KEY INSIGHTS :

- **Top Product Category:**
The **beleza_saude** category generated the highest sales, with total sales of **1,297,490.77**.
- **Top Performing Location:**
São Paulo (SP) recorded the highest sales among the states, with total sales of **5,998,226.96**.
- **Preferred Payment Method:**
Credit card was the payment method generating the highest sales, with total payment value of **12,542,084.19**.
- **Top-Selling Product:**
Product ID **bb50f2e236e5eea0100680137654686c** generated the highest product sales value, totaling **68,485.00**.

- **Best Sales Month:**

November 2017 (2017-11) recorded the highest monthly sales, with total sales of **1,194,882.80**.

CONCLUSION: The analysis helps identify high-performing categories, important geographic markets, customer payment preferences, sales trends, and unusual transactions. These findings can help an e-commerce company improve inventory planning, marketing, pricing, and business decisions.

Question 2 – Student Academic Performance Analysis

A college has Student, Attendance, Internal Marks, Assignment, and Examination datasets. Integrate the datasets and analyze student performance based on department, semester, attendance, and marks. Identify correlations and outliers and visualize performance trends. Provide 5 key findings and recommendations.

AIM: The Student Academic Performance Analysis project analyzes student academic data by integrating information from student records, attendance, internal marks, assignments, and examination results. The analysis helps identify factors affecting academic performance and provides useful insights based on department, semester, attendance, and marks.

DATASETS USED: students, attendance , internal marks , attendance , assignments examination .

TOOLS USED: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook / VS Code.

OVERALL PERFORMANCE ANALYSIS

The total marks were calculated using:

Total Marks = Internal Marks + Assignment Marks + Examination Marks

Average performance was calculated using:

Average Marks = Total Marks / 3

```
===== OVERALL PERFORMANCE =====  
Average Total Marks: 133.04
```

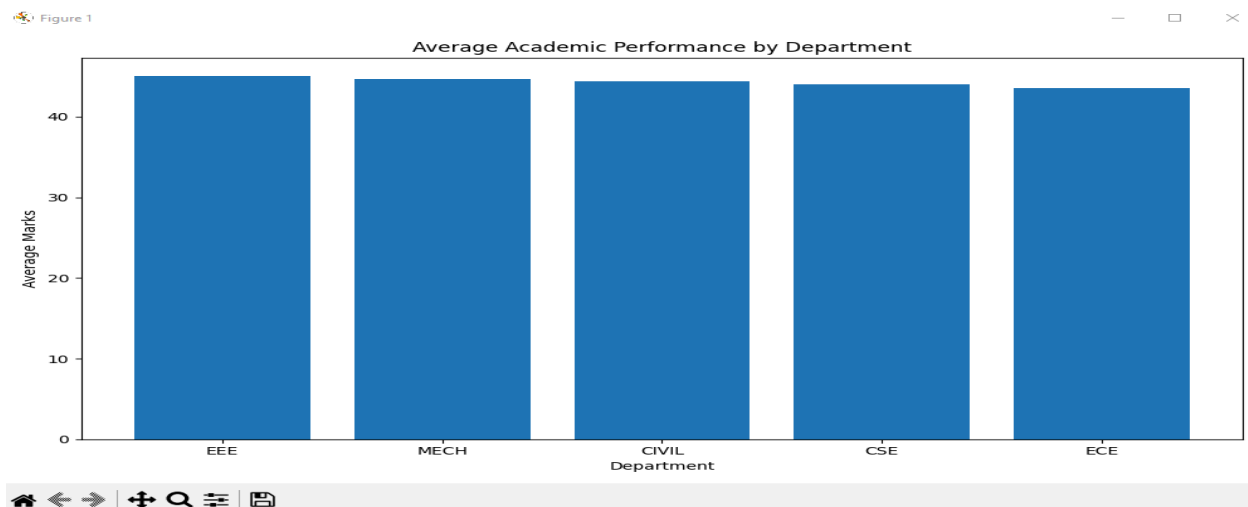
DEPARTMENT ANALYSIS:

Student performance was analyzed according to the department.

The departments included: CSE,ECE,EEE,MECH,CIVIL

The average marks and attendance were calculated for each department.

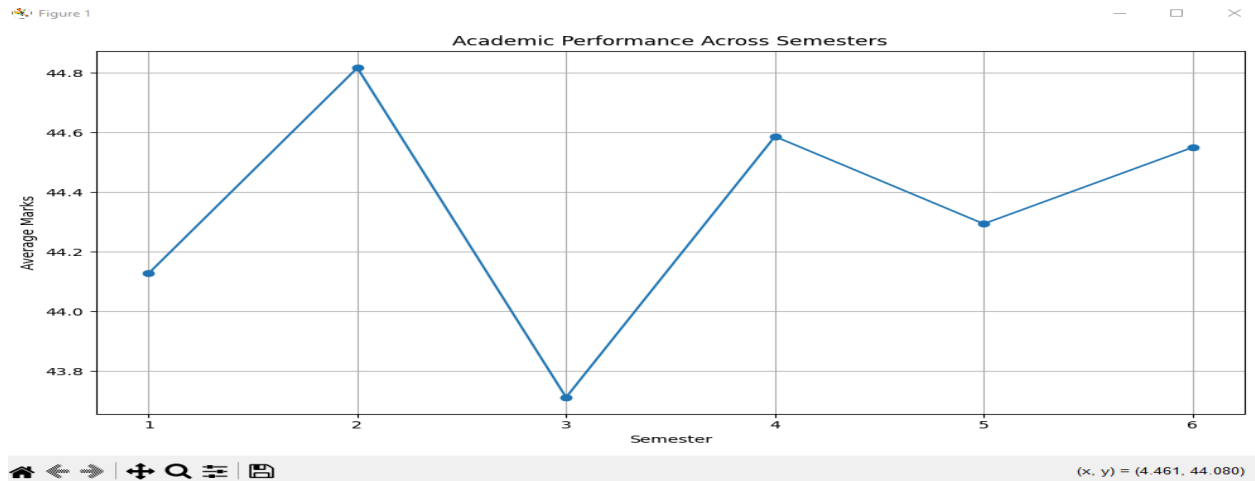
Top 10 Students:					
	student_id	student_name	department	semester	average_marks
331	S0332	Student_0332	ECE	1	63.180000
384	S0385	Student_0385	CIVIL	5	63.026667
305	S0306	Student_0306	MECH	3	62.316667
259	S0260	Student_0260	ECE	6	62.246667
297	S0298	Student_0298	EEE	1	62.080000
358	S0359	Student_0359	CSE	4	62.073333
144	S0145	Student_0145	MECH	2	61.836667
8	S0009	Student_0009	EEE	4	61.716667
341	S0342	Student_0342	MECH	3	61.453333
53	S0054	Student_0054	CIVIL	2	60.903333



SEMESTER ANALYSIS:

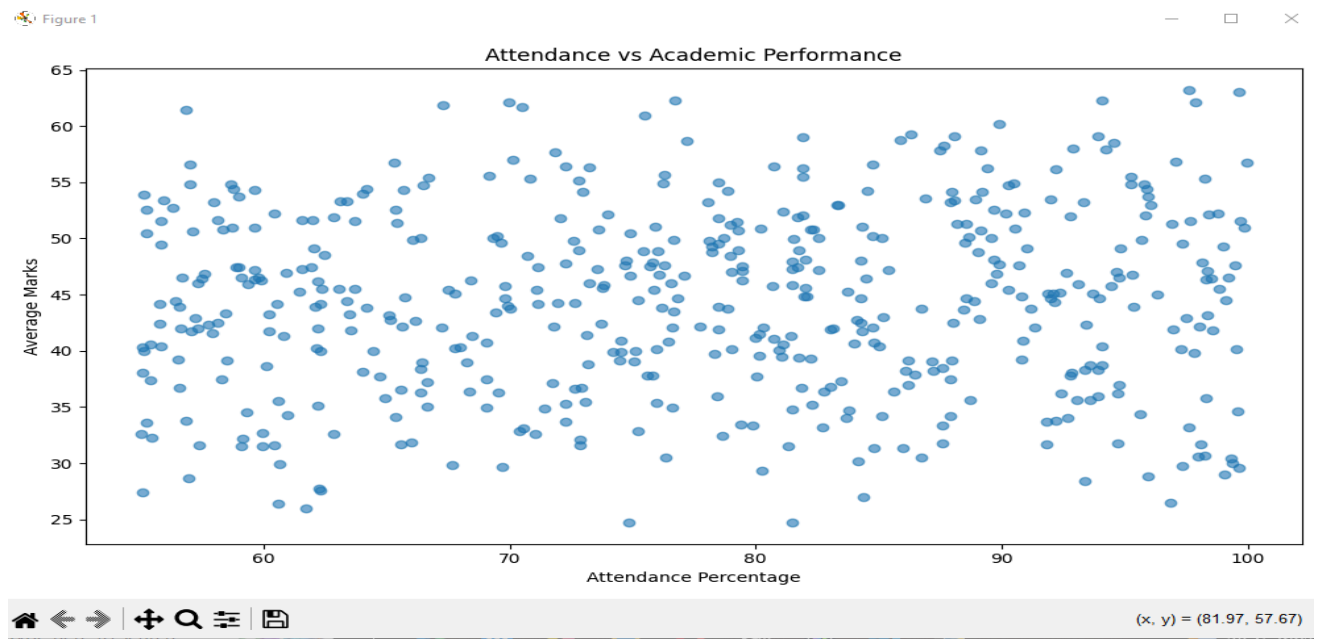
Student performance was compared across semesters from **Semester 1 to Semester 6**.

This helps identify which semester has the highest and lowest average academic performance.



ATTENDANCE ANALYSIS: The relationship between attendance and academic performance was analyzed. Students were grouped into attendance ranges:

- Below 60% ,60–70% ,70–80% ,80–90% ,90–100%.



MARKS ANALYSIS:

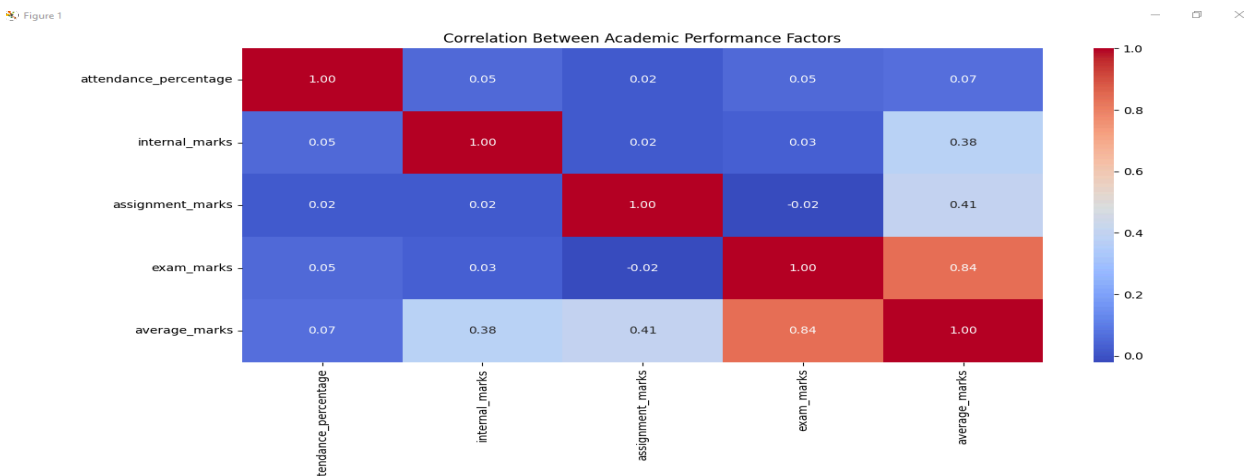
The following academic components were analyzed:

- Internal Marks
- Assignment Marks
- Examination Marks

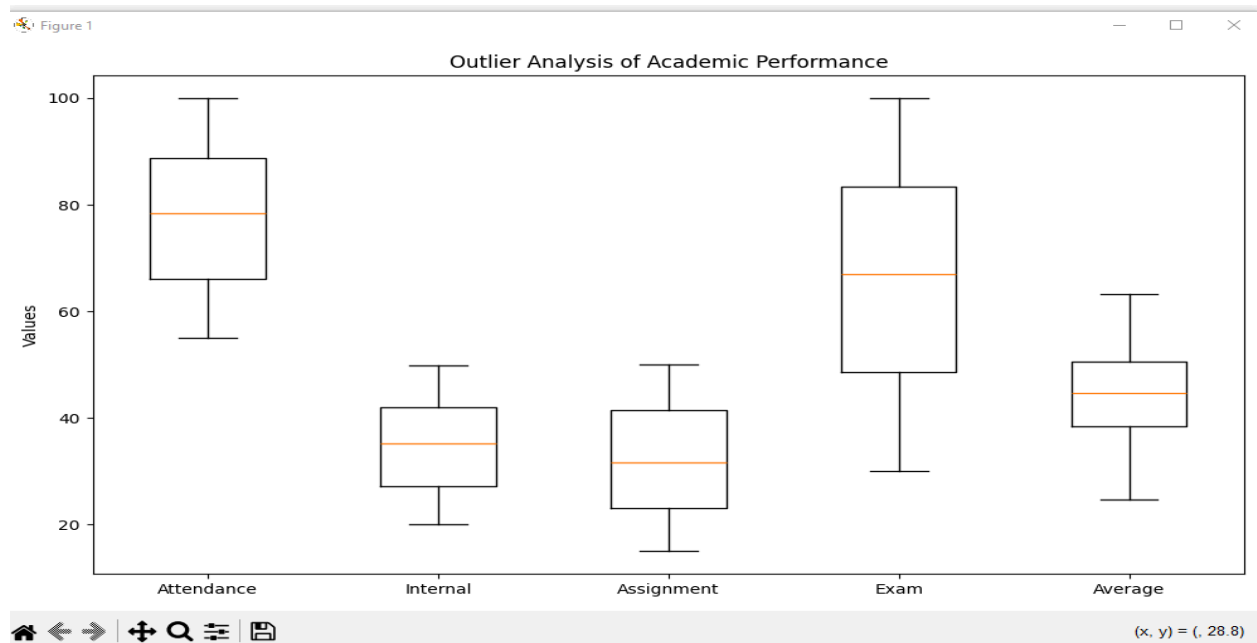
Average scores were calculated to identify the strongest assessment component.

===== ATTENDANCE VS PERFORMANCE =====			
	attendance_percentage	Average_Marks	Number_of_Students
0	Below 60%	44.040404	66
1	60-70%	43.040071	94
2	70-80%	45.010000	109
3	80-90%	44.626033	121
4	90-100%	44.688182	110
===== AVERAGE MARKS =====			
Average Internal Marks: 34.82			
Average Assignment Marks: 32.28			
Average Examination Marks: 65.95			

CORRELATION ANALYSIS :



OUTLIER ANALYSIS :



KEY FINDINGS:

1. Best Performing Department

The EEE department recorded the highest average academic performance, with an average mark of 45.03.

2. Best Performing Semester

Semester 2 recorded the highest average academic performance, with an average mark of 44.82.

3. Attendance and Academic Performance

The correlation between attendance and average marks was 0.07. This indicates a very weak positive relationship between attendance and overall academic performance in this dataset.

4. Strongest Assessment Component

Examination Marks had the highest average score among the assessment components, with an average score of 65.95.

5. Outlier Analysis

Using the IQR (Interquartile Range) method, 0 students were identified as potential outliers in average marks. This indicates that the average-mark values did not contain extreme observations according to this method.

CONCLUSION:

The Student Academic Performance Analysis successfully integrated Student, Attendance, Internal Marks, Assignment, and Examination datasets to study academic performance.

The analysis found that EEE had the highest average departmental performance at 45.03, while Semester 2 had the highest semester performance at 44.82. Examination marks had the highest average score of 65.95. The correlation between attendance and average marks was 0.07, indicating a very weak positive relationship in this dataset. The IQR analysis identified no average-mark outliers.

Overall, the analysis demonstrates how data analytics can help colleges understand academic performance, compare departments and semesters, monitor assessment results, and make informed decisions to improve student outcomes.